

Test of Mathematics for University Admission, 2022 Paper 1

Worked Solutions

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Introduction for students

These solutions are designed to support you as you prepare to take the Test of Mathematics for University Admission. They are intended to help you understand how to answer the questions, and therefore you are strongly encouraged to **attempt the questions first** before looking at these worked solutions. For this reason, each solution starts on a new page, so that you can avoid looking ahead.

The solutions contain much more detail and explanation than you would need to write in the test itself – after all, the test is multiple choice, so no written solutions are needed, and you may be very fluent at some of the steps spelled out here. Nevertheless, doing too much in your head might lead to making unnecessary mistakes, so a healthy balance is a good target!

There may be alternative ways to correctly answer these questions; these are not meant to be ‘definitive’ solutions.

The questions themselves are available at the [TMUA Preparation Materials](#)

Question 1

This equation is a quadratic in $\cos^2 \theta$, and it factorises as

$$(2 \cos^2 \theta - 3)(\cos^2 \theta - 1) = 0.$$

Therefore $2 \cos^4 \theta - 5 \cos^2 \theta + 3 = 0$ if and only if $\cos^2 \theta = \frac{3}{2}$ or $\cos^2 \theta = 1$. The first equation has no real solutions, so the original equation holds if and only if $\cos^2 \theta = 1$, or $\cos \theta = \pm 1$.

$\cos \theta = 1$ has the solutions $\theta = 0$ and $\theta = 2\pi$ in the interval $0 \leq \theta \leq 2\pi$, and $\cos \theta = -1$ has the solution $\theta = \pi$ in this interval.

Therefore the original equation has three real solutions in the interval, and the correct option is C.

Question 2

We rewrite the equation in completed square form: starting from

$$x^2 - 2px + y^2 - 6y - p^2 + 8p + 9 = 0,$$

completing the square for x and y gives

$$(x - p)^2 - p^2 + (y - 3)^2 - 9 - p^2 + 8p + 9 = 0,$$

which simplifies to

$$(x - p)^2 + (y - 3)^2 - 2p^2 + 8p = 0$$

or

$$(x - p)^2 + (y - 3)^2 = 2p^2 - 8p.$$

This is a circle with squared radius $2p^2 - 8p$, as long as this is positive. So the complete set of values of p for which the original equation describes a circle in the xy -plane are those for which $2p^2 - 8p > 0$, or $p^2 - 4p > 0$.

This factorises as $p(p - 4) > 0$, which has roots $p = 0$, $p = 4$, so the complete set of values in which the inequality holds is $p < 0$ or $p > 4$, which is option D.

Question 3

Using the first statement, $f''(x) = a$ for all x , we can integrate to get $f'(x) = ax + b$ for some constant b , and a further integration gives $f(x) = \frac{1}{2}ax^2 + bx + c$ for some constant c .

The second statement then yields $f(0) = c = 1$ and $f(1) = \frac{1}{2}a + b + 1 = 2$, so $a + 2b = 2$.

Finally, we can now use the third statement:

$$\begin{aligned}\int_0^1 f(x) \, dx &= \left[\frac{1}{6}ax^3 + \frac{1}{2}bx^2 + x \right]_0^1 \\ &= \frac{1}{6}a + \frac{1}{2}b + 1 \\ &= 1\end{aligned}$$

so that $a + 3b = 0$.

Combining the simultaneous equations

$$\begin{aligned}a + 2b &= 2 \\ a + 3b &= 0\end{aligned}$$

gives $b = -2$ and $a = 6$, so the correct answer is option F.

Question 4

Suppose that the angle of the sector is θ (in radians). Then the arc length of the smaller sector is $r\theta$ and the area is $\frac{1}{2}r^2\theta$.

For the smaller sector, this gives $r\theta = 6$, so $\theta = 6/r$. Therefore the area is $\frac{1}{2}r^2(6/r) = 3r$.

Likewise, since the larger sector has the same angle as the smaller one, its arc length is $(r+3)(6/r)$ and its area is $\frac{1}{2}(r+3)^2(6/r) = 3(r+3)^2/r$.

We are told the difference between the areas, so we have

$$\frac{3(r+3)^2}{r} - 3r = 21.$$

Multiplying this by r and rearranging gives

$$3(r+3)^2 - 3r^2 - 21r = 0$$

which we can expand and simplify to give

$$-3r + 27 = 0,$$

so $r = 9$.

We can now calculate the arc length of the larger sector; it is $(r+3)(6/r) = 8$.

Therefore, the perimeter of the first sector is $2r + 6 = 24$ and the perimeter of the second sector is $2(r+3) + 8 = 32$. The positive difference is 8, which is option C.

Question 5

We write out the formulae for x_2 , x_3 and x_4 in terms of p and q , using the values $x_1 = 3$, $x_2 = 5$ and $x_3 = 7$:

$$n = 1 : \quad 5 = \frac{3+p}{3+q}$$

$$n = 2 : \quad 7 = \frac{5+p}{5+q}$$

$$n = 3 : \quad x_4 = \frac{7+p}{7+q}$$

Multiplying the first equation by $3+q$ and the second by $5+q$ gives

$$15 + 5q = 3 + p$$

$$35 + 7q = 5 + p$$

These are simultaneous equations for p and q ; subtracting gives $20 + 2q = 2$ so $q = -9$ and therefore $p = -33$.

Substituting these into the expression for x_4 gives

$$x_4 = \frac{7-33}{7-9} = 13$$

so the answer is option H.

Question 6

We calculate the integral and use the laws of logarithms; we obtain

$$\begin{aligned}
 \int_{\log_2 5}^{\log_2 20} x \, dx &= \left[\frac{1}{2} x^2 \right]_{\log_2 5}^{\log_2 20} \\
 &= \frac{1}{2} (\log_2 20)^2 - \frac{1}{2} (\log_2 5)^2 \\
 &= \frac{1}{2} ((\log_2 20)^2 - (\log_2 5)^2) \\
 &= \frac{1}{2} ((\log_2 4 + \log_2 5)^2 - (\log_2 5)^2) \\
 &= \frac{1}{2} ((2 + \log_2 5)^2 - (\log_2 5)^2) \\
 &= \frac{1}{2} (4 + 4 \log_2 5 + (\log_2 5)^2 - (\log_2 5)^2) \\
 &= \frac{1}{2} (4 + 4 \log_2 5) \\
 &= 2 + 2 \log_2 5 \\
 &= \log_2 4 + \log_2 25 \\
 &= \log_2 100
 \end{aligned}$$

so the answer is option F.

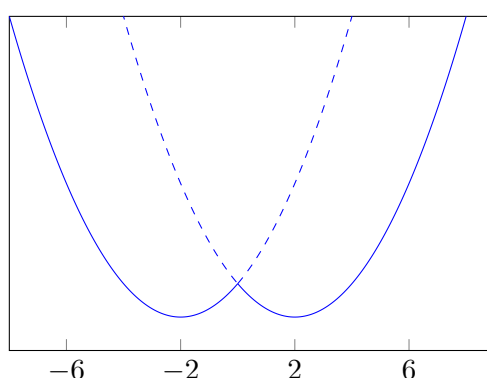
Question 7

The line $y = 0$ is the x -axis; the curve $y = x^2 - 4|x| - 12$ is $y = x^2 - 4x - 12$ when $x \geq 0$ and $y = x^2 + 4x - 12$ when $x < 0$, so we sketch these two graphs.

The curve $y = x^2 - 4x - 12 = (x - 6)(x + 2)$ intersects the x -axis at $x = 6$ and $x = -2$ and has a minimum half way between them, at $x = 2$. Since we are only considering $x \geq 0$, we only observe one point of intersection with the x -axis.

Similarly, the curve $y = x^2 + 4x - 12 = (x + 6)(x - 2)$ intersects the x -axis at $x = -6$ and $x = 2$ and has a minimum half way between them, at $x = -2$. Since we are only considering $x < 0$, we only observe one point of intersection with the x -axis.

We can now sketch the graph, where we have shown the parts of the quadratics we are not using as dashed lines:



The finite area enclosed between the x -axis and the graph $y = x^2 - 4|x| - 12$ is therefore between $x = -6$ and $x = 6$. We could calculate the integrals of the two quadratics, one between $x = -6$ and $x = 0$ and the other between $x = 0$ and $x = 6$. But we could also notice that the graph is symmetrical about the y -axis, so we only need to find one of these and double the answer.

We have

$$\begin{aligned} \int_0^6 x^2 - 4x - 12 \, dx &= \left[\frac{x^3}{3} - 2x^2 - 12x \right]_0^6 \\ &= \frac{6^3}{3} - 2 \times 6^2 - 12 \times 6 - 0 \\ &= -72 \end{aligned}$$

so $\int_{-6}^6 x^2 - 4|x| - 12 \, dx = -144$ and the (unsigned) area required is 144, which is option E.

Question 8

We use the formula $S_n = \frac{a(r^n - 1)}{r - 1}$ and substitute into the given equation:

$$\frac{a(r^{30} - 1)}{r - 1} - \frac{a(r^{20} - 1)}{r - 1} = k \frac{a(r^{10} - 1)}{r - 1}.$$

We can multiply this by $r - 1$ and divide by a to get

$$(r^{30} - 1) - (r^{20} - 1) = k(r^{10} - 1),$$

which simplifies to

$$r^{30} - r^{20} = k(r^{10} - 1).$$

Factorising the left hand side then gives

$$r^{20}(r^{10} - 1) = k(r^{10} - 1);$$

as $r > 1$, we can divide by $r^{10} - 1 \neq 0$ to obtain

$$r^{20} = k.$$

Therefore k is a 20th power, and the smallest possible value is 2^{20} , which is option B.

Question 9

It may well require reading this question a couple of times to understand what it is asking! It seems like we need to find all possible functions $f(x)$ and $g(x)$, then for each one, to find the minimum value that $f(x)$ takes, and then to find the minimum of all the minima. There might, of course, be some shortcuts!

Let us try to solve the equations to find $f(x)$. The first equation rearranges to give $g(x) = f(x) - 2 \sin x$, so we can substitute this into the second equation to give

$$f(x)(f(x) - 2 \sin x) = \cos^2 x.$$

This is a quadratic in $f(x)$, so we can rearrange it into the standard form and use the quadratic formula:

$$(f(x))^2 - (2 \sin x)f(x) - \cos^2 x = 0,$$

hence

$$\begin{aligned} f(x) &= \frac{2 \sin x \pm \sqrt{4 \sin^2 x + 4 \cos^2 x}}{2} \\ &= \sin x \pm 1. \end{aligned}$$

So there are two possible functions $f(x)$, namely $f(x) = \sin x + 1$ and $f(x) = \sin x - 1$. The minimum value taken by these for any x is 0 for the first function and -2 for the second, so the minimum possible value is -2 , which is option E.

Question 10

Any sequence of translations is equivalent to a translation in the x -direction followed by a translation in the y -direction (where the translations could be zero).

Translating $y = x^3$ by a in the x -direction gives the graph

$$y = (x - a)^3 = x^3 - 3ax^2 + 3a^2x - a^3.$$

Then translating by b in the y -direction gives the graph

$$y = x^3 - 3ax^2 + 3a^2x - a^3 + b.$$

Now considering the three graphs:

- I The x^2 term is $-3x^2$, which means we require $a = 1$. Substituting this into the above general equation gives $y = x^3 - 3x^2 + 3x - 1 + b$, which does not match the given equation. So this is not a translation of $y = x^3$.
- II The x^2 term is $-9x^2$, so we require $a = 3$. Substituting gives $y = x^3 - 9x^2 + 27x - 27 + b$, which matches the equation with $b = 24$, so this is a translation of $y = x^3$.
- III The x^3 term does not match here, so this is not a translation of $y = x^3$.

Therefore the answer is option C, II only.

Question 11

We have $\log_{10}(3^{1-n}) = (1-n)\log_{10} 3$, so the sum becomes

$$\begin{aligned}\sum_{n=1}^{100} \log_{10}(3^{1-n}) &= \sum_{n=1}^{100} (1-n)\log_{10} 3 \\ &= (\log_{10} 3) \sum_{n=1}^{100} (1-n).\end{aligned}$$

Now the sum is actually an arithmetic series: expanding it gives

$$\begin{aligned}\sum_{n=1}^{100} (1-n) &= 0 + (-1) + (-2) + \cdots + (-99) \\ &= -(0 + 1 + 2 + \cdots + 99) \\ &= -\frac{1}{2} \times 100(0 + 99 \times 1) \\ &= -4950\end{aligned}$$

using the formula $S_n = \frac{1}{2}n(2a + (n-1)d)$. (Alternatively, we could have used the formula for the n th triangular number.)

Therefore the original sum evaluates to $-4950\log_{10} 3$, which is option A.

Question 12

The formula for y_k is given in completed square form. The minimum of the graph $y = a(x-b)^2+c$, where $a > 0$, is at $x = b$, and the minimum value is c . Therefore the minimum of y_k is at $(\frac{k}{2}, \frac{k^2}{2} + 4k + 3)$.

The minimum of the minimum y -coordinates is therefore the minimum of $\frac{k^2}{2} + 4k + 3$. We can find this minimum either by completing the square or by differentiating this with respect to k . Let us use the latter method here as it seems simpler. We have

$$\frac{d}{dk} \left(\frac{k^2}{2} + 4k + 3 \right) = k + 4$$

so the minimum occurs at $k = -4$, giving

$$\left(\frac{k}{2}, \frac{k^2}{2} + 4k + 3 \right) = (-2, -5)$$

so that $a + b = -7$, which is option D.

Question 13

Expanding the brackets gives

$$2 - a^3b^3 + \frac{4}{a^3b^3} - 2 = \sqrt{2}$$

so

$$\frac{4}{a^3b^3} - a^3b^3 = \sqrt{2}.$$

We are interested in the least possible value of ab , so we can find the least possible value of a^3b^3 and take the cube root. Let us simplify the notation by writing $x = a^3b^3$, so $\frac{4}{x} - x = \sqrt{2}$.

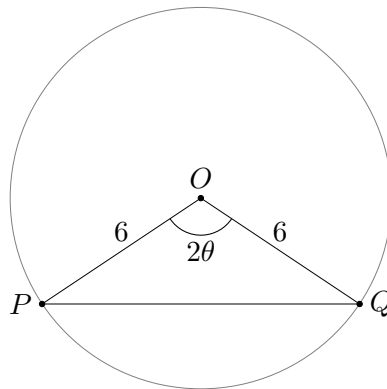
Multiplying by x gives a quadratic: $4 - x^2 = \sqrt{2}x$, which we can rearrange to $x^2 + \sqrt{2}x - 4 = 0$ and then solve using the quadratic formula:

$$x = \frac{-\sqrt{2} \pm \sqrt{2+16}}{2} = \frac{-\sqrt{2} \pm 3\sqrt{2}}{2} = -2\sqrt{2} \text{ or } \sqrt{2}.$$

Since we want the least possible value of x , we take $x = a^3b^3 = -2\sqrt{2} = -2^{\frac{3}{2}}$, and so the minimum possible value of ab is $(-2^{\frac{3}{2}})^{\frac{1}{3}} = -2^{\frac{1}{2}} = -\sqrt{2}$, which is option A.

Question 14

Let us start by drawing the triangle POQ in the circle:



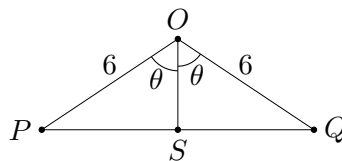
(We have called the angle 2θ as we will shortly want to consider half of the angle.)

We are told that the area of POQ is $9\sqrt{3}$, so we can use the formula for the area of a triangle given two sides and the included angle to determine 2θ : the area formula is $A = \frac{1}{2}ab \sin C$, which in this case gives

$$\frac{1}{2} \times 6 \times 6 \times \sin 2\theta = 9\sqrt{3}$$

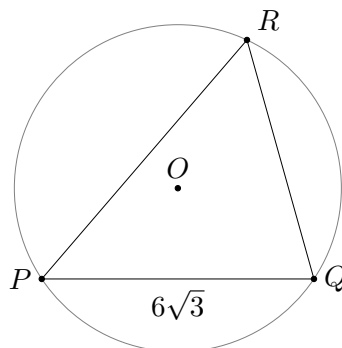
so $\sin 2\theta = \frac{\sqrt{3}}{2}$. Since $2\theta \geq \frac{\pi}{2}$, we must have $2\theta = \frac{2}{3}\pi$ (or 120°).

We can now work out the length PQ by dropping a perpendicular from O to PQ :

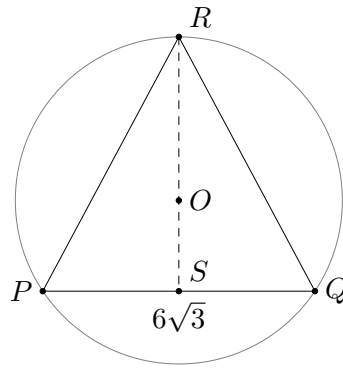


We have $PS = SQ = 6 \sin \theta = 6 \sin \frac{1}{3}\pi = 3\sqrt{3}$, so $PQ = 6\sqrt{3}$, and we also have $OS = 6 \cos \theta = 6 \cos \frac{1}{3}\pi = 3$.

We can now work out the maximum possible area of the triangle PQR . The triangle PQR looks like this:



Fixing PQ and allowing R to move, we see that the largest possible area of PQR occurs when R is as far as possible from PQ , as the area of the triangle is $\frac{1}{2} \times \text{base} \times \text{height}$. When R is at the top of the circle (in our diagram), we have this configuration:



We have $RS = RO + OS$. Now $RO = 6$ (it is a radius of the circle) and $OS = 3$ as we calculated earlier, so $RS = 9$. Therefore the maximum possible area of PQR is

$$\frac{1}{2} \times PQ \times RS = \frac{1}{2} \times 6\sqrt{3} \times 9 = 27\sqrt{3},$$

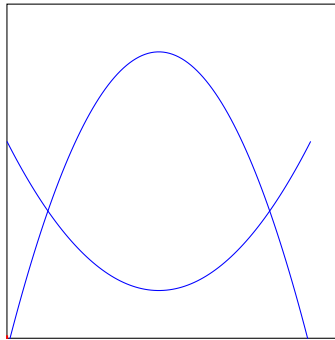
which is option D.

Question 15

We start by sketching the two curves to understand what is happening.

They are both quadratics. The graph of p has a maximum at $(0, 8)$ and the graph of q has a minimum at $(0, -2)$. They are both symmetrical about $x = 0$ (the y -axis). It might be helpful to work out where they intersect: $8 - 2x^2 = x^2 - 2$ implies $3x^2 = 10$, so $x = \sqrt{10/3}$, which is between 1.5 and 2.

The graph of these therefore looks like this, where we have also sketched a possible rectangle. For the rectangle to have the maximum possible area, it must meet the graphs at all four of its vertices.



Let us suppose that the vertices of the rectangle are at $x = \pm a$, with $0 < a < \sqrt{10/3}$. Then the width of the rectangle is $2a$ and the height is $(8 - 2a^2) - (a^2 - 2) = 10 - 3a^2$. Therefore the area is $A = 2a(10 - 3a^2) = 20a - 6a^3$.

To maximise this area, we differentiate with respect to a to find the stationary points. We have

$$\frac{dA}{da} = 20 - 18a^2$$

so there is a stationary point when this is zero, at $a = \pm\sqrt{20/18} = \pm\sqrt{10/9} = \pm\frac{1}{3}\sqrt{10}$. We are only interested in the positive value, and this is less than $\sqrt{10/3}$, so it is a valid value for our problem. We therefore obtain the maximum possible area: it is

$$\begin{aligned} A &= 2a(10 - 3a^2) \\ &= \frac{2}{3}\sqrt{10}(10 - \frac{1}{3} \times 10) \\ &= \frac{2}{3}\sqrt{10} \times \frac{2}{3} \times 10 \\ &= \frac{40\sqrt{10}}{9} \end{aligned}$$

which is option H.

Question 16

The given quartic can be thought of as a quadratic in x^2 , and as such it has roots

$$x^2 = \frac{6 \pm \sqrt{6^2 - 28}}{14} = \frac{3 \pm \sqrt{2}}{7}.$$

Since the roots of the quartic are $\pm \cos \theta$ and $\pm \cos \beta$, these x^2 values are the values of $\cos^2 \theta$ and $\cos^2 \beta$ (in some order).

Now $\sin^2 \theta = 1 - \cos^2 \theta$ and $\sin^2 \beta = 1 - \cos^2 \beta$, so

$$\sin^2 \theta, \sin^2 \beta = 1 - \frac{3 \pm \sqrt{2}}{7} = \frac{4 \mp \sqrt{2}}{7}.$$

Therefore the equations $x^2 - \frac{4+\sqrt{2}}{7} = 0$ and $x^2 - \frac{4-\sqrt{2}}{7} = 0$ have solutions $\pm \sin \theta$ and $\pm \sin \beta$ (again, in some order, and x is not the same as earlier). It follows that the product of these two equations will have the four solutions $\pm \sin \theta$ and $\pm \sin \beta$; this equation is

$$\left(x^2 - \frac{4+\sqrt{2}}{7}\right)\left(x^2 - \frac{4-\sqrt{2}}{7}\right) = 0.$$

To obtain the form given in the question, we first expand the brackets:

$$x^4 - \left(\frac{4+\sqrt{2}}{7} + \frac{4-\sqrt{2}}{7}\right)x^2 + \frac{4+\sqrt{2}}{7} \cdot \frac{4-\sqrt{2}}{7} = 0$$

then perform the surd calculations to give

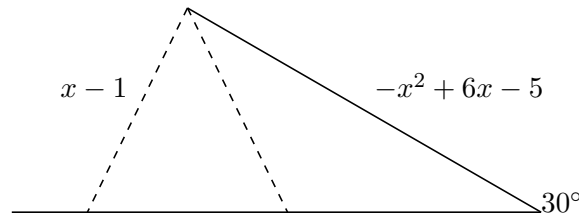
$$x^4 - \frac{8}{7}x^2 + \frac{2}{7} = 0.$$

Finally, multiplying by 7 gives $7x^4 - 8x^2 + 2 = 0$, which is option B.

Question 17

We first require that the lengths shown are both positive; $x - 1 > 0$ if and only if $x > 1$, and $-x^2 + 6x - 5 = -(x - 5)(x - 1) > 0$ if and only if $1 < x < 5$. Therefore we require $1 < x < 5$ to ensure both lengths are positive.

There are two non-congruent triangles when the side marked $x - 1$ can be drawn in two configurations:



This is possible when the following conditions are satisfied:

- $x - 1 < -x^2 + 6x - 5$ (otherwise one of the possibilities will lie to the right of the 30° angle in the above diagram)
- $x - 1 > (-x^2 + 6x - 5) \sin 30^\circ$ (so that the distance from the end of the $-x^2 + 6x - 5$ side to the base is at least $x - 1$, making it possible to form a triangle; if there is equality here, then the dashed circle will be tangent to the base of the triangle, and there will only be one possibility)

The first inequality rearranges as $x^2 - 5x + 4 < 0$, which factorises as $(x - 1)(x - 4) < 0$, giving $1 < x < 4$.

The second equality, using $\sin 30^\circ = \frac{1}{2}$, becomes $2x - 2 > -x^2 + 6x - 5$, so $x^2 - 4x + 3 > 0$. This factorises as $(x - 1)(x - 3) > 0$, so $x < 1$ or $x > 3$.

Requiring both of these inequalities to hold thus gives $3 < x < 4$; this also satisfies the positivity requirement we started with ($1 < x < 5$), and so the answer is option D.

Question 18

The equation $f(x) = g(x)$ is a polynomial equation with degree 5, so has at most 5 distinct real solutions; we wish there to be 5 such solutions for all values of p .

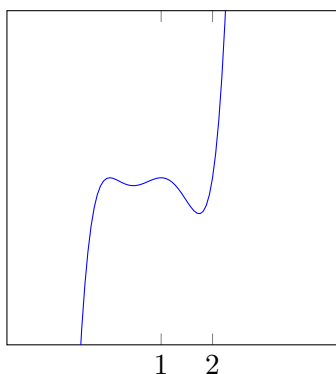
We would have fewer than 5 distinct real solutions if we either have a square factor of the form $(x - a)^2$ for some a , or if the graph of $y = f(x) - g(x)$ has fewer than four turning (stationary) points, or if this graph has a minimum point above the x -axis or a maximum point below it, for then it would not cross the x -axis 5 times.

Let us write

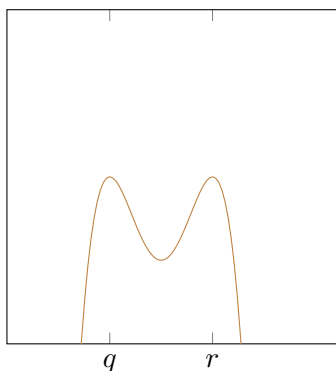
$$h(x) = f(x) - g(x) = x^2(x - 1)^2(x - 2) + p(x - q)^2(x - r)^2$$

We can try expanding and differentiating this, but that looks horrible. Let us think about the above description, though, remembering that there need to be 5 distinct real solutions for *all* values of p .

The graph of $y = f(x) = x^2(x - 1)^2(x - 2)$ intersects the x -axis at $x = 0$ (where it is tangent to the axis, because it is a double root), $x = 1$ (likewise) and $x = 2$, and it is positive when $x > 2$ and negative or zero when $x < 2$, so the graph looks like this:

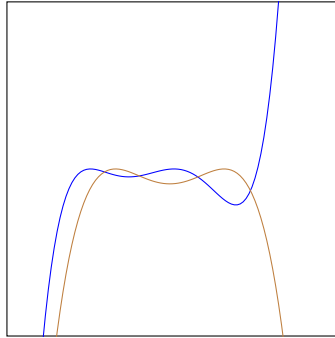


The graph of $y = g(x) = -p(x - q)^2(x - r)^2$ has two points of intersection with the x -axis, one at $x = q$ and one at $x = r$, and at both of these points, the graph is tangent to the axis. (In the case $q = r$, there is just one point of tangency.) The graph is entirely below or touching the x -axis as we are given that $p > 0$. It looks like this:



If $r > 2$, then the graph of $y = g(x)$ will entirely miss one of the “valleys” of $y = f(x)$ if p is large, as it will if $r < 1$; in this case, there would be at most three points of intersection for

large p . So we must have $1 < r < 2$. If $1 < q < r$, then it will miss the left valley for large enough p , so we must have $0 < q < 1$ and $1 < r < 2$. In this case, the graph of $y = g(x)$ must intersect both valleys twice as $g(1) < 0$:



This sketch shows four points of intersection, and we are looking for five. But the graph of $y = f(x)$ tends to $-\infty$ as x tends to $-\infty$ far faster than $y = g(x)$, regardless of the value of p , as $y = f(x)$ is a quintic but $y = g(x)$ is only a quartic. So there will always be another point of intersection with $x < 0$.

Hence the correct answer is option B.

Note that there is actually a small simplification in the above argument. We showed that we cannot have $1 < q < r$ and deduced that we must have $0 < q < 1$. But what about the possibility $q = 1$? In that case, there would be a double root at $x = 1$, and so there would not be five distinct roots, hence we cannot have $q = 1$. But what about r ? We showed that we cannot have $r > 2$ or $r < 1$. If $r = 1$, we have the same problem, but what about $r = 2$ and $0 < q < 1$? In this case, it is still true that the equation has five distinct real solutions for all positive p , as $x = 2$ is only a single root of $f(x) = 0$. So the full set is actually $q < 1$ and $1 < r \leq 2$.

Question 19

Note first that the region where the centre of C_2 may lie is entirely within the circle C_1 , as $2^2 + 3^2 = 13 < 25$.

As the radius of C_1 is 5, the circle C_2 will intersect C_1 if and only if the centre of C_2 is at least a distance of 1 from the origin (the centre of C_1). This is because in this case, a line from the centre of C_1 passing through the centre of C_2 will reach the circle C_1 before (or as) it reaches the circle C_2 , but a line from the centre of C_1 in the opposite direction will reach C_2 first. If, though, the centre of C_2 is at a distance of less than 1 from the origin, the furthest point on C_2 from the origin will be at a distance of less than $1 + 4 = 5$ from the origin, so C_2 lies entirely inside C_1 .

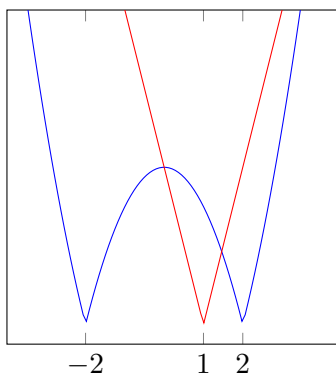
The area of the allowed (a, b) region is $4 \times 6 = 24$. The (a, b) region where the circles do not intersect is a circle of radius 1, with area π , lying entirely within the allowed region, so the region where the circles do intersect has area $24 - \pi$.

Thus the probability of intersection is $\frac{24 - \pi}{24}$, which is option F.

Question 20

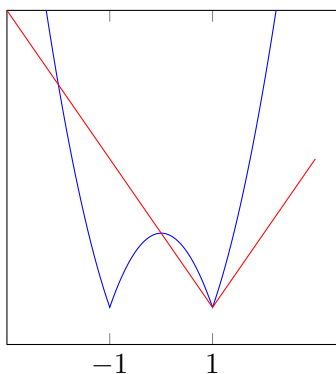
The graph $y = |x^2 - a^2|$ intersects the x -axis at $\pm a$. The graph $y = a^2|x - 1|$ intersects the x -axis at $x = 1$. Replacing a with $-a$ does not change the equations, so we may as well assume that $a \geq 0$.

Here is a sketch of the graphs with $a = 2$; we note that the graphs always intersect at $x = 0$ (and $y = a^2$ there):



We immediately see that when $a > 1$, there will be 4 points of intersection: the graph $y = a^2|x - 1|$ is below the “hill” at $(1, 0)$, so intersects it twice as x moves away from $x = 1$, and then intersects the quadratic $y = x^2 - a^2$ for some $x < -a$ and some $x > a$ as the quadratic grows faster than the linear $y = -a^2(x - 1)$ or $y = a^2(x - 1)$ in these regions.

What happens when $a = 1$? In this case, the equations become $y = |x^2 - 1|$ and $y = |x - 1|$. The derivative of the quadratic $y = x^2 - 1$ is $2x$, so the gradient is just over 2 when x is just over 1. (There is no gradient at the point $x = 1$.) Therefore the graph looks like this:

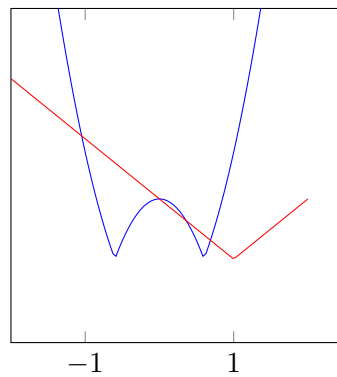


We see that we have lost one point of intersection: there are now only three points of intersection.

What happens if $0 < a < 1$? Now the vertex of $y = a^2|x - 1|$ will be to the right of $x = a$, where the quadratic meets the axis, and so there will be no points of intersection there – the quadratic grows too quickly to meet the linear function for $x > 1$. (To be more precise: the gradient of the quadratic to the right of $x = a$ is $2x > 2a > a^2$, so the quadratic will be growing faster than the linear function.)

On the left of $x = a$, we have one point of intersection to the left of $x = -a$ as before. There is also a point of intersection at $(0, a^2)$. Since this is the vertex of the quadratic, the linear

function must go below the “hill”, giving another two points of intersection, yielding four points of intersection in total:



There is one remaining case to consider: $a = 0$. In this case, the graphs become $y = |x^2| = x^2$ and $y = 0$, and these graphs have a single point of intersection at the origin.

Therefore the possible numbers of points of intersection are 1, 3 and 4, and the smallest number not possible is $n = 2$, option B.

Test of Mathematics for University Admission, 2022 Paper 2

Worked Solutions

Version 1.0, July 2023

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Introduction for students

These solutions are designed to support you as you prepare to take the Test of Mathematics for University Admission. They are intended to help you understand how to answer the questions, and therefore you are strongly encouraged to **attempt the questions first** before looking at these worked solutions. For this reason, each solution starts on a new page, so that you can avoid looking ahead.

The solutions contain much more detail and explanation than you would need to write in the test itself – after all, the test is multiple choice, so no written solutions are needed, and you may be very fluent at some of the steps spelled out here. Nevertheless, doing too much in your head might lead to making unnecessary mistakes, so a healthy balance is a good target!

There may be alternative ways to correctly answer these questions; these are not meant to be ‘definitive’ solutions.

The questions themselves are available on the ‘**TMUA Resources**’

Question 1

We differentiate to obtain

$$\frac{dy}{dx} = 12x^3 + 12x^2 + 12x.$$

We solve the equation $\frac{dy}{dx} = 0$ to find the number of stationary points:

$$\begin{aligned} 12x^3 + 12x^2 + 12x &= 0 \\ \text{if and only if} \quad 12x(x^2 + x + 1) &= 0. \end{aligned}$$

So either $x = 0$ or $x^2 + x + 1 = 0$. But this quadratic has discriminant $1^2 - 4 \times 1 = -3 < 0$, so it has no real solutions.

Thus there is only one stationary point, at $x = 0$, and the answer is option B.

Question 2

We expand the brackets in the first term using the binomial theorem and write out the sum in full to get

$$(1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5) \times (1 + x + x^2 + x^3 + x^4 + x^5).$$

The x^5 term in the product is obtained by taking all possible pairs of a term in the first bracket and a term in the second whose product is cx^5 for some number c , and then adding these together. Therefore the x^5 term in the product is

$$1 \cdot x^5 + 5x \cdot x^4 + 10x^2 \cdot x^3 + 10x^3 \cdot x^2 + 5x^4 \cdot x + x^5 \cdot 1 = (1 + 5 + 10 + 10 + 5 + 1)x^5 = 32x^5.$$

Therefore the correct answer is option E.

Question 3

- I $n = 2$ is prime and $n^2 + 2 = 6$ is not prime, so this case satisfies the statement and is not a counterexample.
- II $n = 3$ is prime but $n^2 + 2 = 11$ is prime, so this is a counterexample as it does not satisfy the statement.
- III $n = 4$ is not prime, so it satisfies the statement and is not a counterexample.

Therefore only II provides a counterexample, which is option C.

Question 4

We can rewrite the equation of circle by completing the square to find its centre. We get:

$$(x + f)^2 - f^2 + (y + g)^2 - g^2 + h = 0$$

which we can rearrange as

$$(x + f)^2 + (y + g)^2 = f^2 + g^2 - h.$$

Therefore the centre is at $(-f, -g)$ (and the radius is $\sqrt{f^2 + g^2 - h}$, but that is not relevant to us here).

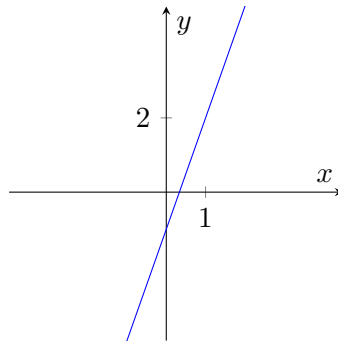
Then the distance between the centre of the circle and the point P is given by

$$L^2 = (p + f)^2 + (q + g)^2$$

so we need the values of f , g , p and q . Hence the correct answer is option B.

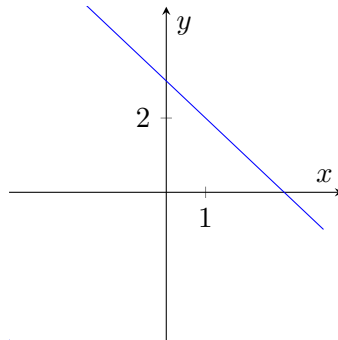
Question 5

- I Let us sketch a line through $(1, 2)$ with a negative y -intercept:



It is clear from this sketch that if the line has a negative y -intercept, it must have a positive x -intercept. (A more formal proof goes as follows: since $y < 0$ when $x = 0$ and $y > 0$ when $x = 1$, there must be some value of x with $0 < x < 1$ where $y = 0$. Therefore the x -intercept must lie between 0 and 1.)

- II The converse of P reads: If the x -intercept of L is positive, then the y -intercept of L is negative. This is the case if the x -intercept is between 0 and 1, as in the previous sketch, but if the x -intercept is greater than 1, this is no longer true:



So the converse of P is false.

- III The contrapositive of P has the same truth value as P itself, so the contrapositive of P is true.

Therefore only I and III are true, and the correct answer is option F.

Question 6

Suppose P is true. Since there are an odd number of integers in the list, there is a unique middle one (when they are written in increasing order), so this is the median. Therefore Q is true, and P is sufficient for Q.

Now suppose that Q is true. It does not follow that n is odd; here is a counterexample: if the list is 2, 2, then $n = 2$ and the median is 2, so Q is true, but P is false. (In general, any list with n even and the middle pair of numbers equal is a counterexample.) Therefore P is not necessary for Q.

The correct answer is option C.

Question 7

Searching for small counterexamples to the claim does not find any; the first few differences are all prime. So we do have to check the proof in detail, as it might be correct.

The algebraic expansion on line I is correct.

The expression in line II is also correct, using the expansion from line I.

For line III, if we could write $3x^2 + 3x + 1 = (ax + b)(cx + d)$ with real coefficients a, b, c and d , then we would have real roots of the equation $3x^3 + 3x + 1 = 0$, namely $x = -\frac{b}{a}$ and $x = -\frac{d}{c}$. But the discriminant is $3^2 - 4 \times 3 \times 1 = -3 < 0$, so this is impossible. So this statement is true.

Line IV is more tricky: just because the algebraic expression is not factorisable into algebraic factors does *not* mean that the integer it represents for any particular value of x is not factorisable in integers. As a simpler example, the algebraic expression $x^2 + 1$ is not factorisable into real algebraic factors, yet substituting integers for x shows that the integer value can sometimes be factorised: taking $x = 3$ gives $3^2 + 1 = 10 = 2 \times 5$, for example. So this line of the argument is not valid.

Line V does follow from lines II and IV.

Therefore the proof is wrong, and the first (and only) error occurs on line IV (option F).

It is worth noting that the ‘opposite’ argument does work: if an algebraic expression does factorise as $(ax + b)(cx + d)$ with a, b, c and d integers, then the value of original expression always factorises (as an integer) when x is an integer, just by substituting the value of x into the factorised form. (It is possible, though, that one of the factors is 1, so the integer may be prime for some value(s) of x .)

It also turns out that in this case, if we search far enough, we do find a counterexample to the claim: $6^3 - 5^3 = 216 - 125 = 91 = 7 \times 13$. (It might have been the case that the claim was true even though the attempted proof had an error.)

Question 8

We first determine the number of terms in the whole sequence: there are $\frac{70-1}{3} + 1 = 23 + 1 = 24$ terms.

We next determine the number of pairs that sum to 74. The pairs are:

$$\begin{array}{c} 4 + 70 \\ 7 + 67 \\ \vdots \\ 34 + 40 \\ [37 + 37] \end{array}$$

The final pair is excluded because the two terms have to be distinct to fit the rule in (*). The number of valid pairs is therefore $\frac{34-4}{3} + 1 = 10 + 1 = 11$.

If a selection S does *not* satisfy the rule in (*), then it can have at most one term from each of these pairs, so it has to leave out at least 11 terms. So it can have at most $24 - 11 = 13$ terms. Therefore if S has at least 14 terms, it must satisfy (*).

We also need to show that if S only has 13 terms, it might not satisfy (*). We can include 1, 37 and one term from each listed pair, giving $2 + 11 = 13$ terms. These 13 terms do not contain a distinct pair summing to 37.

Hence the smallest value of n which forces (*) to be true is 14, and the correct answer is option C..

Question 9

The statement (*) does not restrict us to *positive* real numbers x , so whatever the value of k , we can always find a large negative number x with $x < k$ and $x^2 \geq k$.

We can even be explicit about this (though that is not required to answer this question): if $k \leq 0$, we can take $x = k - 1$ in which case $x^2 > 0 \geq k$, and if $k > 0$, we can take $x = -\sqrt{k}$ so $x < 0 < k$ and $x^2 = k \geq k$.

Therefore (*) is true for no real values of k , and the correct option is A.

Question 10

- I We can find a counterexample to this statement: taking $x = 2$ and $n = 1$ shows that this statement is false.
- II This is true; if $x < 0$, we can take $n = 1$, and if $x > 0$, we can take n to be the smallest integer greater than x .
- III This is true; we can take $x = 0$. (If, though, we replaced ‘positive integers’ with ‘integers’, the resulting statement would be false.)

Therefore II and III are true, and the correct option is G.

Question 11

The angle SPQ is a right angle if and only if the triangle SPQ satisfies Pythagoras's theorem, that is, if and only if

$$SP^2 + PQ^2 = SQ^2.$$

Since $PQRS$ is a kite, its diagonals intersect at right angles, so both SOP and POQ are right-angled triangles. Therefore, again by Pythagoras,

$$\begin{aligned} SP^2 &= x^2 + z^2 \\ PQ^2 &= x^2 + y^2. \end{aligned}$$

Substituting this, along with $SQ = y + z$, into the above equation gives the following necessary and sufficient condition for angle SPQ to be a right angle:

$$x^2 + z^2 + x^2 + y^2 = (y + z)^2.$$

Expanding and simplifying, this is equivalent to

$$2x^2 = 2yz$$

or $x^2 = yz$.

Therefore option C gives a necessary and sufficient condition.

(One should also check that none of the others are necessary and sufficient; we could take $x = 2$, $y = 4$, $z = 1$: this satisfies option C but none of the others.)

Question 12

We do not know how to integrate any of these, so we will instead compare the integrands as the integrals are all from 0 to 1.

We have $(\sqrt{2})^x = 2^{x/2}$, which makes R look a little simpler. Since $x/2 < x$ for $0 < x < 1$, $2^{x/2} < 2^x$ in this interval, and so $R < Q$.

Now in this interval, we also have $x < \sqrt{x}$ (as $x^2 < x$), so $2^x < 2^{\sqrt{x}}$, hence $Q < P$.

Combining these, we find that $R < Q < P$, so the correct option is F.

Question 13

For each fixed value of x , we can think about this expression as being a function of y . We can write the expression as $(1 - x)y + x$, which is just $my + c$ where $m = 1 - x$ and $c = x$. This is the equation of a straight line graph, so it takes every real value, both positive and negative, as long as $m \neq 0$.

The case $m = 0$ occurs exactly when $x = 1$, and in this case, the expression becomes $(1 - 1)y + 1 = 1$, so it is always 1, for every value of y .

Therefore (*) is true for all except exactly one value of x , namely $x = 1$, and the correct answer is option E.

Question 14

To solve this question, we use the fact that $|x - a|$ can be understood as the distance of x from a .

The first inequality is true if and only if x is closer to -5 than to -11 , i.e., if and only if $x > -8$.

The second inequality is true if and only if x is closer to -11 than to -1 , i.e., if and only if $x < -6$.

Therefore both inequalities are true if and only if $-8 < x < -6$, which is an interval of length 2, which is option D.

One could also do this question by sketching the graphs of $y = |x + 5|$ and $y = |x + 11|$ and working out where they intersect, and so on. But that requires a lot more work than the approach presented here.

Question 15

Since $\log_x y = z$, we have $y = x^z$. Likewise $z = y^x$. Combining these gives

$$z = y^x = (x^z)^x = x^{xz}.$$

Therefore raising the equation to the power of $\frac{1}{xz}$ gives $z^{1/xz} = x$, hence $\log_z x = \frac{1}{xz}$, which is option F.

Question 16

For each of these, we can try to construct a counterexample, and if that doesn't work, we may understand why the statement must be true.

For simplicity in the explanations below, let's write

$$A_{\min} = \text{minimum of } a_1, \dots, a_{100}$$

$$B_{\min} = \text{minimum of } b_1, \dots, b_{100}$$

$$C_{\min} = \text{minimum of } c_1, \dots, c_{100}$$

and similarly for $A_{\max}$ etc.

- I Let's suppose that $B_{\min} = 0$ and $C_{\min} = 0$. Can we then make the minimum of the a_n s greater than zero?

Yes, we can: if we take $b_1 = 0$ and the rest of the b_n s to be 100, and we take $c_{100} = 0$ and the rest of the c_n s to be 100, then we can have $a_n = 100$ for every n , so the minimum of the a_n s is 100.

Therefore this statement is not necessarily true.

- II If we try the same example as in I, we find that $A_{\min} = 100$, which is greater than $B_{\min} + C_{\min}$, so the inequality holds in this case.

But to get a counterexample, we want to make $A_{\min}$ *small*. Helpfully, the condition $a_n \leq b_n + c_n$ allows us to make each a_n as small as we like. So if we take $a_n = 0$, $b_n = 1$ and $c_n = 1$ for all n , then the condition will be satisfied for each n , but we will have $A_{\min} = 0$ and $B_{\min} = C_{\min} = 1$, so $A_{\min} < B_{\min} + C_{\min}$ in this case.

Therefore this statement is not necessarily true either.

- III Both of the counterexamples we used for I and II satisfy this statement. It is not obvious how to construct a counterexample: every simple example satisfies this statement. So let us instead try to prove it.

Each a_n satisfies the condition $a_n \leq b_n + c_n$. Now $b_n \leq B_{\max}$ and $c_n \leq C_{\max}$ for each n , so $a_n \leq B_{\max} + C_{\max}$ for each n . But this means that the maximum of all of the a_n s also satisfies this condition, that is $A_{\max} \leq B_{\max} + C_{\max}$, so the given statement must be true.

Therefore only statement III must be true, and the correct answer is option D.

Question 17

Considering the offered options, the focus is on the order of the steps and whether the steps prove what they should be proving; there is no requirement for us to check the algebraic calculations themselves.

The task is to prove that **if** $27b \left(b + \frac{4a^3}{27} \right) < 0$, **then** $x^3 + ax^2 + b = 0$ has three distinct real roots.

In step I, the student finds the stationary points of $y = x^3 + ax^2 + b$.

In step II, the student assumes that $27b \left(b + \frac{4a^3}{27} \right) < 0$, which is the correct thing to do to prove an ‘if ... then’ statement.

In step III, the student says ‘**if** the cubic has three distinct real roots **then** ...’, which is correct but is not useful: we know that one of the stationary points is above the x -axis and the other is below and we wish to deduce that the cubic has three distinct real roots.

In step IV, the student uses the result of II and the result ‘**if** one of the stationary points is above the x -axis and one is below, **then** the equation has three distinct real roots’, which is the converse of the statement in step III.

Therefore the correct option is E: the student should have shown the converse of the result in step III.

Question 18

Let us substitute values into the four equations to begin with.

When $x = 0$, we have

$$\begin{aligned}(\cos x)^{\cos x} &= 1^1 = 1 \\(\sin x)^{\sin x} &= 0^0 = ? \\(\cos x)^{\sin x} &= 1^0 = 1 \\(\sin x)^{\cos x} &= 0^1 = 0\end{aligned}$$

This shows that graph Q is $y = (\sin x)^{\cos x}$. (We do not know what $(\sin x)^{\sin x}$ is when $x = 0$, but from the graphs given, since three of the functions have value 1 at $x = 0$ and only one is 0, it must be 1.)

Next, let us consider the value at $x = \frac{\pi}{2}$:

$$\begin{aligned}(\cos x)^{\cos x} &= 0^0 = ? \\(\sin x)^{\sin x} &= 1^1 = 1 \\(\cos x)^{\sin x} &= 0^1 = 0 \\(\sin x)^{\cos x} &= 1^0 = 1\end{aligned}$$

Again, as only one graph has value 0 at $x = \frac{\pi}{2}$, graph P depicts $(\cos x)^{\sin x}$. (We did not actually need to calculate the final line of this list, as we have already identified this function as being graph Q.)

So we are left with $(\cos x)^{\cos x}$ and $(\sin x)^{\sin x}$ as graphs R and S in some order. Let us put in the value $x = \frac{\pi}{6}$, as that is clearly different between the two functions. We have

$$\begin{aligned}(\cos x)^{\cos x} &= \left(\frac{\sqrt{3}}{2}\right)^{\frac{\sqrt{3}}{2}} \\(\sin x)^{\sin x} &= \left(\frac{1}{2}\right)^{\frac{1}{2}}\end{aligned}$$

The first of these looks horrible, but we can approximate the value of the second of these; it is just $\frac{1}{\sqrt{2}} \approx 0.7$, so this must be graph S. Similarly, $(\cos x)^{\cos x} \approx 0.7$ at $x = \frac{\pi}{3}$, which is graph R.

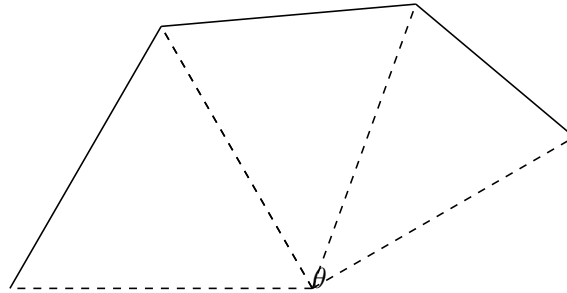
Therefore the graphs are:

$$\begin{aligned}(\cos x)^{\cos x} &: \text{graph R} \\(\sin x)^{\sin x} &: \text{graph S} \\(\cos x)^{\sin x} &: \text{graph P} \\(\sin x)^{\cos x} &: \text{graph Q}\end{aligned}$$

which is option E.

Question 19

We sketch a part of the polygon with the triangles described in the question:



Suppose the angle of one of the triangles is θ , as shown. Then the area of the triangle is given by $\frac{1}{2}r^2 \sin \theta$ where r is the radius of the circle (using the formula $A = \frac{1}{2}ab \sin C$).

Since r is the same for each of the triangles, all the triangles will have equal area if and only if $\sin \theta$ is the same for each triangle. This will certainly be the case if the polygon is regular, but we have to determine if this can be true in any other situation.

If two of the triangles have angles at the centre of the circle of θ and ϕ , then $\sin \phi = \sin \theta$ if and only if $\phi = \theta$ or $\phi = \pi - \theta$. Can we have the latter possibility? If so, it would mean we have some angles being θ and some being $\pi - \theta$.

Suppose then that θ is acute and $\pi - \theta$ is obtuse. (We cannot have a reflex angle as the centre of the circle lies inside the polygon, and if θ is a right angle, then $\pi - \theta = \theta$.) Suppose further that k of the angles of the polygon triangles equal $\pi - \theta$ and the remaining $n - k$ equal θ . Then the sum of these triangle angles is given by

$$(n - k)\theta + k(\pi - \theta) = 2\pi$$

since they form a whole circle. We can rearrange this to $(n - 2k)\theta = (2 - k)\pi$ and hence

$$\theta = \frac{(2 - k)\pi}{n - 2k}.$$

There can be at most three obtuse angled triangles (as four would be more than a whole circle), so let's try $k = 1$, $k = 2$ and $k = 3$ in turn.

If $k = 1$, we have $\theta = \frac{\pi}{n-2}$. It looks as though this will work for every n , but we have to be a little careful as we need θ to be acute. When $n = 3$, we get $\theta = \pi$, which is not allowed. When $n = 4$, we get $\theta = \frac{\pi}{2}$, which is again not allowed. When $n > 4$, $\theta < \frac{\pi}{2}$, so this does work. Therefore when $n \geq 5$, it is possible for the conditions to be satisfied but for the polygon not to be regular.

Now consider $k = 2$, giving $\theta = \frac{0}{n-4}$. We only need to consider $n = 3$ and $n = 4$, as we have dealt with $n \geq 5$ above. This is zero when $n = 3$, so it does not work for a triangle. When $n = 4$, this formula is not meaningful (it gives $\frac{0}{0}$). We can instead go back to the original equation $(n - 2k)\theta = (2 - k)\pi$, which becomes $0 = 0$ in this case, meaning that it works for any value of θ . And indeed, this gives two angles of θ and two of $\pi - \theta$, so we get a non-regular polygon with equal area triangles. The simplest example is a non-square rectangle.

We are left with the case $k = 3$ to consider. When $n = 3$, there would be no acute angles and three obtuse ones, and the formula gives $\theta = \frac{-\pi}{-3} = \frac{\pi}{3}$. And indeed, a triangle with three

obtuse angles of $\frac{\pi}{3}$ is an equilateral triangle satisfying the conditions. Therefore the only triangle ($n = 3$) satisfying the conditions is regular.

We have thus been able to construct non-regular polygons satisfying the conditions for every $n \geq 4$, but not for $n = 3$. Hence the correct answer is option B.

Question 20

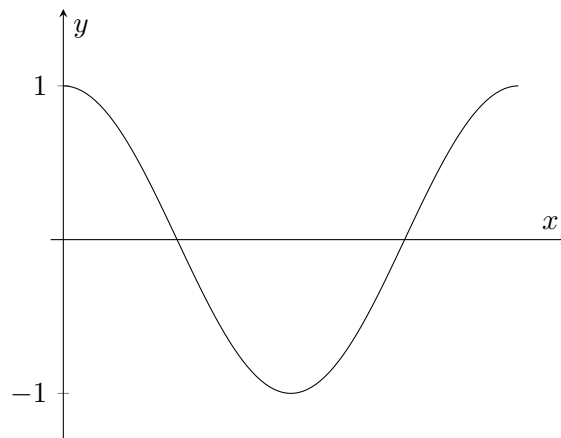
We note that we can rewrite each of the functions f_2 to f_5 in terms of the previous function:

$$\begin{aligned} f_1(x) &= \cos x \\ f_2(x) &= \sin f_1(x) \\ f_3(x) &= \cos f_2(x) \\ f_4(x) &= \sin f_3(x) \\ f_5(x) &= \cos f_4(x) \end{aligned}$$

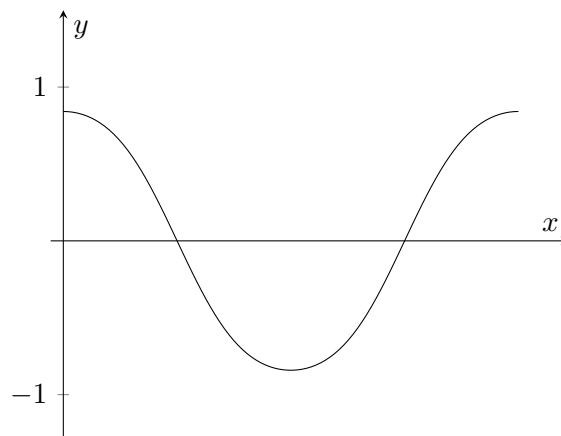
Since $f_1(x) = \cos x$ is periodic with period 2π , each of the other functions also repeats every 2π (though it may have a shortest period less than this), and so we only need to consider $0 \leq x \leq 2\pi$. (We could actually restrict ourselves to $0 \leq x \leq \pi$, as the values of $\cos x$ in the range $\pi < x \leq 2\pi$ are a repeat of the values in the range $0 \leq x \leq \pi$, but we will draw a complete period anyway.)

Sketching the graph of each of these functions is the most straightforward way to proceed. We note first a result that we will need later: $\pi \approx 3.14$ so $\frac{\pi}{2} \approx 1.6$ and hence $\sin 1 < \sin \frac{\pi}{2} = 1$.

A sketch of $f_1(x) = \cos x$ is straightforward:

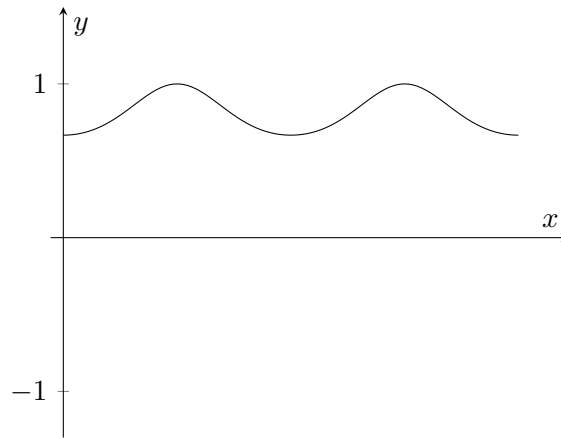


We see that $m_1 = 1$, and the minimum value taken is -1 . Now $f_2(x) = \sin f_1(x)$, so this goes between $-\sin 1$ and $\sin 1$, with roots where $f_1(x) = 0$; recall that $\sin 1 < 1$:



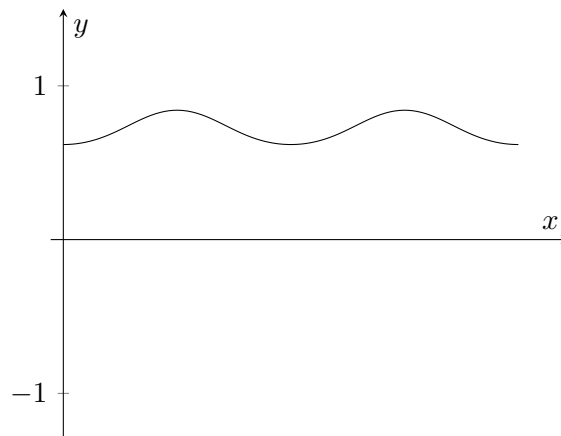
We see that $m_2 = \sin 1 < 1$, and the minimum value taken is $-\sin 1$.

Now $f_3(x) = \cos f_2(x)$. When $f_2(x) = 0$, $f_3(x) = 1$, so $f_3(x)$ has a maximum of 1 at $x = \frac{\pi}{2}$ and at $x = \frac{3\pi}{2}$. But $f_3(x)$ never reaches 0 as $f_2(x)$ never reaches $\frac{\pi}{2} \approx 1.6$ or $-\frac{\pi}{2} \approx -1.6$. Therefore the graph looks something like this:



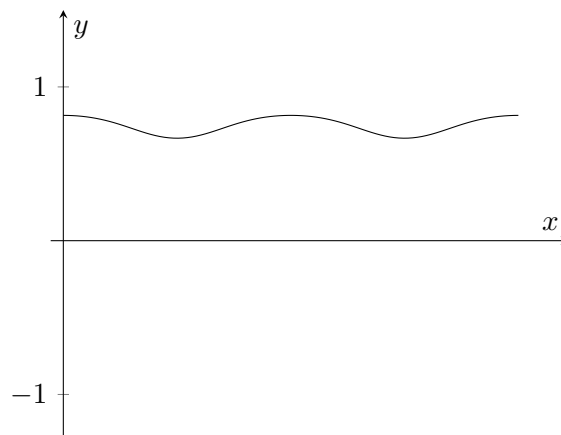
Therefore $m_3 = 1$, and the minimum value taken is $\cos(\sin 1)$.

Next, $f_4(x) = \sin f_3(x)$, so this will be an oscillating curve going between $\sin(\cos(\sin 1))$ and $\sin 1$:



Thus $m_4 = \sin 1$.

Finally, $f_5(x) = \cos f_4(x)$, so this will be an oscillating curve going between $\cos(\sin 1)$ and $\cos(\sin(\cos(\sin 1)))$:



To summarise, we have $m_1 = m_3 = 1$, $m_2 = m_4 = \sin 1$ and $m_5 = \cos(\sin(\cos(\sin 1)))$, and the correct option is E.

It is also interesting to note from these accurate graphs that the subsequent functions are getting flatter and flatter and seem to be tending to a straight line (though the limiting value is different in the functions f_n where n is even and those where n is odd).